

Mohnish Shende

Cybersecurity and infrastructure engineer | Nagpur, India

mohnish@mohnish.ink | mohnish.ink | github.com/MohnishShende | linkedin.com/in/mohnishshende

PROFESSIONAL SUMMARY

Cybersecurity and infrastructure engineer working across vulnerability assessment, Linux-based infrastructure, Python security tooling, OSINT research, and technical reporting. Builds practical tools and documentation for scanning workflows, self-hosted systems, secure configuration, and structured security analysis. Brings an evidence-led research style to technical work, with emphasis on traceability, bounded conclusions, and usable documentation.

PROFESSIONAL EXPERIENCE

Project Associate I

CSIR - National Environmental Engineering Research Institute | Nagpur, India | Dec 2024 - Mar 2025

- Conducted vulnerability assessments using Nessus and Burp Suite.
- Supported network segmentation and access-control policy implementation.
- Documented secure-configuration baselines and system-hardening guidelines.
- Prepared technical reports for security assessments and infrastructure-hardening activities.

SELECTED TECHNICAL WORK

nmap-llm - open-source security tooling

Python, Nmap, Ollama, Mistral, XML, JSON, Markdown | github.com/MohnishShende/nmap-llm

Offline working prototype for Nmap planning, execution, XML parsing, and local-LLM-assisted triage reporting.

- Built Python modules for configuration, local model interaction, scanning, parsing, and utilities.
- Converted Nmap XML into structured data and Markdown-oriented reporting workflows.
- Used local Ollama/model integration to keep assistance within an offline workflow.
- No automated test suite was observed; safety and adversarial-input behavior require further validation.

NodeZero - infrastructure engineering and recovery assessment

Debian, Linux, Docker Compose, Caddy, DNS, Tailscale, Samba, SSH, nftables, systemd | github.com/MohnishShende/NodeZero

Public-safe documentation and assessment of a self-hosted Linux environment across service boundaries, security posture, and recovery readiness.

- Produced a structured engineering handbook with architecture, security, and recovery sections.
- Applied normalized inventories, architecture reconstruction, bounded findings, and traceable recommendations.
- Distinguished configuration presence from operational proof, particularly for recovery.
- Owner-produced work, not independent client assurance; topology and operational evidence remain private.

ECI EVM Engineering Assurance - claim-level technical assessment

Evidence registers, claim review, traceability, methodology | github.com/MohnishShende/ECI-EVM-Engineering-Assurance

Structured evaluation of public evidence concerning the ECI EVM ecosystem.

- Organized source records, claim-review packages, traceability, conclusions, and explicit exclusions.
- Repository reports a substantive v1 release; SSRN publication/posting remains independently unconfirmed.
- Firmware, source code, hardware teardown, and live election audit remain outside scope.

SECURITY RESEARCH AND TECHNICAL REPORTING

Bounded OSINT reconnaissance methodology

OSINT, passive and semi-passive reconnaissance, evidence handling, severity writing

Developed a structured public-source reconnaissance and reporting method with explicit scope, positive controls, confidence, risk notes, and remediation guidance.

- Sensitive target observations and actionable source material remain private.
- No confirmed CERT-In submission or response is claimed.

PEER-REVIEWED PUBLICATION

Psychological Appropriation: A Dimensional Model of Surrogate-Moral Violence

Journal of Forensic Psychology Research and Practice | Published online 5 Dec 2025 | DOI 10.1080/24732850.2025.2599908

Conceptual dimensional model examining surrogate identification, moralized logic, symbolic consent, affective idealization, and narrative ritualization. The theoretical model requires empirical validation.

SELECTED PREPRINTS

Covert Thought Reform: Adapting Classical Models Across Political, Cultural, and Organizational Contexts

SSRN preprint | DOI 10.2139/ssrn.5452614

Comparative historical analysis adapting classical influence models to proposed covert variants.

Belief, Faith, and the Misclassification of Religious Commitment

SSRN preprint | DOI 10.2139/ssrn.6447879

Conceptual analysis distinguishing belief, disbelief, suspension, acceptance, and faith.

The Inescapable Chain: Human Nature, Exploitation, and the Illusion of Freedom

SSRN preprint | DOI 10.2139/ssrn.5364398

Interdisciplinary argumentative synthesis across philosophy, history, and psychology.

TECHNICAL SKILLS

Security: Nmap, Nessus, Burp Suite, command allowlisting, scan-result parsing, OSINT, passive reconnaissance

Infrastructure: Linux, Debian, Docker, Docker Compose, Caddy, Pi-hole, Unbound, Tailscale, Samba, SSH, nftables, systemd

Development: Python, Flask, REST APIs, XML, JSON, Markdown, requests, xmldict, tracemalloc

Research and documentation: technical reporting, source verification, evidence evaluation, confidence labeling, chronology reconstruction

EDUCATION

MSc Cyber Security

University of Plymouth | Merit | Awarded 11 Nov 2024

NCSC-certified Master's programme in Cyber Security; academic year 2023-2024.

BE Information Technology

Rashtrasant Tukadoji Maharaj Nagpur University | Distinction | 2022

RESEARCH IDENTITIES

Google Scholar: scholar.google.com/citations?user=Qu1HDUIAAAAJ&hl=en

ORCID: 0009-0000-9466-3056 | SSRN author ID: 7734216